

Wei Fang

AI SCIENTIST

✉ weifang@csail.mit.edu

🏠 people.csail.mit.edu/weifang

📄 wei-fang-tw

🌐 Wei Fang

Education

Massachusetts Institute of Technology

Cambridge, MA, USA

PH.D. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

2026

- Advisor: Dr. James R. Glass
- Thesis: Towards Scalable and Generalizable Agentic Large Language Models.

S.M. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

2021

National Taiwan University

Taipei, Taiwan

B.S. IN ELECTRICAL ENGINEERING

2018

Research Focus

Agentic AI: enhancing LLM agents' tool-using capabilities and efficiency with post-training and inference-time learning approaches

Reliable Generation: improving generation factuality with retrieval and knowledge grounding

Experience

Subconscious

Cambridge, MA, USA

FOUNDING AI SCIENTIST

Mar. 2026 (expected)

- I will be joining Subconscious to build efficient, long-context LLM agents.

MIT CSAIL Spoken Language Systems Group

Cambridge, MA, USA

RESEARCH ASSISTANT, SUPERVISED BY DR. JAMES R. GLASS

Sep. 2018 - Mar. 2026

- **Agentic Tool-using** Designed a zero-shot framework for tool instruction tuning to enhance LLM tool-using [ACL25], and a lightweight query-planning module for SOTA multi-step tool retrieval [Preprint].
- **Information Retrieval** Proposed a joint re-ranking method based on pointwise mutual information for improved passage re-ranking [EACL24], and query expansion ranking to boost open-domain QA [ACL23].
- **Factuality & Verification** Factual and fair language generation with self-verification [ASRU23W]; Automatic fact-checking system [NAACL19]; Stance detection with multi-task transfer learning [FEVER19].
- **Other** conversational QA [ACL23W]; temporal information extraction [Preprint]; speech synthesis [Preprint].

TEACHING ASSISTANT

Feb. 2021 - May 2021

- 6.864 Advanced Natural Language Processing

MIT-IBM Watson AI Lab

Cambridge, MA, USA

RESEARCH INTERN

Jun. 2024 - Aug. 2024

- Collaboration with MIT-IBM researchers on the tool instruction optimization project [ACL25].

Apple Inc. Frameworks NLP Team

Cupertino, CA, USA

NLP INTERN

Jul. 2018 - Aug. 2018

- Developed a synthetic data generation pipeline using generative models for the QuickPath project.

NLP INTERN

Jun. 2017 - Sep. 2017

- Investigated multilingual input representations for neural models to enhance user keyboard experience.

NTU & Academia Sinica Vision & Learning Lab

Taipei, Taiwan

RESEARCH ASSISTANT, SUPERVISED BY PROF. YU-CHIANG FRANK WANG

Jul. 2016 - Jan. 2018

- Incorporated knowledge graphs for multi-label zero-shot image recognition [CVPR18].

- Proposed a hierarchical attention-based model for the TOEFL Listening Comprehension Test [SLT16] [TASLP19].
- Researched on unsupervised audio word embeddings.

- EE5177/EE5184 Machine Learning; CSIE5610 Data Analytics & Modeling; EE2011 Signals and Systems

Publications

- [1] **Wei Fang**, James Glass. Beyond Single-Shot: Multi-step Tool Retrieval via Query Planning. *arXiv Preprint*, 2026.
- [2] **Wei Fang**, Yang Zhang, Kaizhi Qian, James Glass, Yada Zhu. PLAY2PROMPT: Zero-shot Tool Instruction Optimization for LLM Agents via Tool Play. *Findings of the Association for Computational Linguistics: ACL*, 2025.
- [3] **Wei Fang**, Yung-Sung Chuang, James Glass. Joint Inference of Retrieval and Generation for Passage Re-ranking. *Findings of the Association for Computational Linguistics: EACL*, 2024.
- [4] Tianhua Zhang, Hongyin Luo, Yung-Sung Chuang, **Wei Fang**, Luc Gaitskell, Thomas Hartvigsen, Xixin Wu, Danny Fox, Helen Meng, James Glass. Interpretable Unified Language Checking. *Symposium for Celebrating 40 Years of Bayesian Learning in Speech and Language Processing and Beyond at ASRU*, 2023.
- [5] Yung-Sung Chuang, **Wei Fang**, Shang-Wen Li, Wen-tau Yih, James Glass. Expand, Rerank, and Retrieve: Query Reranking for Open-domain Question Answering. *Findings of the Association for Computational Linguistics: ACL*, 2023.
- [6] Tianhua Zhang, Liping Tang, **Wei Fang**, Hongyin Luo, Xixin Wu, Helen Meng, James Glass. ConvRGX: Recognition, Generation, and Extraction for Self-trained Conversational Question Answering. *DialDoc Workshop on Document-grounded Dialogue and Conversational Question Answering at ACL*, 2023.
- [7] **Wei Fang** and James Glass. Cross-Document Temporal Relation Extraction with Temporal Anchoring Events. *Preprint*, 2021.
- [8] **Wei Fang**. On End-to-end Automatic Fact-checking Systems. *S.M. Thesis*, 2021.
- [9] **Wei Fang**, Moin Nadeem, Mitra Mohtarami, James Glass. Neural Multi-task Learning for Stance Prediction. *EMNLP-IJCNLP Workshop on Fact Extraction and VERification*, 2019.
- [10] **Wei Fang**, Yu-An Chung, James Glass. Towards Transfer Learning for End-to-end Speech Synthesis from Deep Pre-trained Language Models. *arXiv Preprint*, 2019.
- [11] Chia-Hsuan Lee, Hung-Yi Lee, Szu-Lin Wu, Chi-Liang Liu, **Wei Fang**, Jui-Yang Hsu, Bo-Hsiang Tseng. Machine Comprehension of Spoken Content: TOEFL Listening Test and Spoken SQuAD. *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLPRO)*, 2019.
- [12] Moin Nadeem, **Wei Fang**, Brian Xu, Mitra Mohtarami, James Glass. FAKTA: An Automatic End-to-End Fact Checking System. *Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019.
- [13] Chung-Wei Lee[†], **Wei Fang**[†], Chih-Kuan Yeh, Yu-Chiang Frank Wang. Multi-Label Zero-Shot Learning with Structured Knowledge Graphs. *IEEE Conference on Computer Vision & Pattern Recognition (CVPR)*, 2018.
- [14] **Wei Fang**[†], Jui-Yang Hsu[†], Hung-Yi Lee, Lin-Shan Lee. Hierarchical Attention Model for Improved Comprehension of Spoken Content. *IEEE Workshop on Spoken Language Technology (SLT)*, 2016.

[†] indicates equal contribution

Honors

Government Scholarship to Study Abroad, Ministry of Education, Taiwan

May 2021

Student Travel Bursary, Amazon

Oct. 2019

EECS Great Educators Fellowship, MIT EECS

Feb. 2018

Conference Grant, Foundation for the Advancement of Outstanding Scholarship

Dec. 2016

Professional Activity

Reviewing ACL Rolling Review (2021-), ACL (2021-), EMNLP (2021-), NeurIPS (2024-), ICML (2025-)
Affiliations Sigma Xi, the Scientific Research Honor Society (2025-)

Skills

Languages Python, Bash/Shell
ML Toolkits PyTorch, HuggingFace (Transformers, TRL), vLLM, verifiers (RL)
ML Methodologies SFT, RL, RAG, Agentic, Data Generation
Tools Git, WandB, Slurm, $\LaTeX$